Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
PI Name:	Morris, Jeffrey F. Ph.D.		
Project Title:	Droplet formation processes from particle-laden liquids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031	**Congressional District:**	4
Comments:	Professor F. Joseph Schork has acted as PI at Georgia Tech, while Dr. Jeff Morris (who was at Halliburton) remained technically responsible for the project. The grant is being transferred to CCNY where Dr Morris is joining the faculty January 1, 2005.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/01/2002	**End Date:**	01/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Nahra, Henry K	**Contact Phone:**	216-433-5385
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Flight Program:			
Flight Assignment:	NOTE: End date renegotiated to 1/31/2007, per PI (3/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The study addresses basic drop formation processes for the dripping (or pendant drop) mode and laminar jet instability for solids-laden liquids, processes previously intensively studied for pure liquids. This work is building from that knowledge base by experimentally examining drop formation processes of noncolloidal suspensions at particle volume fractions up to, to date, 40% and eventually 50%. The tools used include high-speed video imaging and image analysis, with analytical tools from suspension mechanics and stability of fluid flow applied to the interpretation of the behavior. The primary issues are to determine the role of particle fraction and particle size, relative to the orifice diameter, in the drop formation process. The study is designed to isolate bulk rheological effects from finite particle size effects, in support of the goals of establishing the similarities and differences of the drop formation process and the associated fluid mechanical instabilities in a suspension of solid particles in relation to the equivalent processes in pure liquids. The basic physical processes in drop formation from mixtures are unexplored, while the phenomena are found in a number of applications. Thus the significance includes fundamental development of suspension mechanics in a free-surface flow with known instabilities, and also will provide guidance to the development of technologies for solid application through drop location processes. The most prevalent of these are inkjet processes in the electronics and textile industries, where the goal is to apply solids in fine lines for circuitry and imaging, respectively. Particle-laden "inks" raise issues in both cases because the very rapid pinch of small drops leads to a situation where the influences of rheology and isolated particle effects are largely unknown. Increased knowledge thus promises to open the way to new scientific approaches and at the same time the elucidation of basic processes should rapidly aid in the design of inkjet applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The benefits of the research toward technologies on earth can be expected in coating and inkjet type of applications. The basic science of mixture flows with free surfaces is generally poor, and is especially so when there is a rupture event such as drop formation involved: the presence of the particles renders the process significantly irregular and thus requires not only a reexamination using suspension mechanical concepts, but also statistical concepts. The advances made here will support understanding of inkjet printing with, for example, metal-laden liquids or ceramic particles, technologies under examination (largely by empirical approaches) for a range of applications.
Task Progress:	The overall task goal is to characterize pendangt In the past year, the work supported by the NASA funding has established the basic parameter dependence of the regimes seen in pendant drop formation and slow jetting, and this work has been published. The work suggests a "two-stage" process in the development of a pendant drop in a liquid containing suspended solids, even at small solid volume fraction but with effects most pronounced as this loading increases. From this work, further examination of the detailed behavior in the "necking" region in the formation of a drop at an orifice was done. The dynamics of this evolution process were experimentally determined for a range of solids fractions and ratio of particle-to-orifice diameters. This showed the validity of the two-stage process for the description of the pendant drop formation, with the early (far from thread rupture) stage agreeing well with the assumption that the particles affect the process only by increasing the effective viscosity, but the latter stage (near to and including rupture) very different than a continuum fluid mechanical event. It is observed that rather than the neck decaying more slowly for a suspension (than the pure liquid counterpart), there is instead a more rapid pinching to rupture for the suspension than a pure liquid, with the resulting length of the thread between the orifice and forming drop achieving much longer lengths for the pure fluid. This effect, which is extremely pronounced as the solid fraction becomes large, is due to the appearance of a localized pinching as the system finds a weak point in the neck. This work has been submitted for publication. Certain results of the above have been shown to be related to the behavior of rebounding drops laden with solids as they impact a solid surface, a problem critical in application of coatings containing solids. This work was also published in the past year.
Bibliography Type:	Description: (Last Updated: 06/25/2007)
Articles in Peer-reviewed Journals	Furbank RJ, Morris JF. "An experimental study of particle effects on drop formation." Physics of Fluids 2004 May;16(5):1777-90. http://dx.doi.org/10.1063/1.1691034 , May-2004
Articles in Peer-reviewed Journals	Ok H, Park H, Carr WW, Morris JF, Zhu JY. "Particle-laden drop impacting on solid surfaces." Journal of Dispersion Science and Technology 2004;25(4):449-56. http://dx.doi.org/10.1081/DIS-200025704 , Jul-2004
Articles in Peer-reviewed Journals	Furbank, R. J.; Morris, J. F. "Pendant drop thread dynamics of particle-laden liquids." Int. J. Multiphase Flow, submitted December 2004. , Dec-2004
Dissertations and Theses	Furbank RJ. "Drop formation from particulate suspensions." PhD thesis, Georgia Institute of Technology, May 2004. , May-2004